

Retail Sales Data Processing and Business Insights Generation

1. Objective

The objective of this project is to clean and analyze retail sales data and generate business insights using Python and Power BI.

2. Dataset Used

Three datasets were used:

- retail_data1
- retail_data2
- product_details

3. Data Cleaning Steps

The following steps were performed:

- Combined retail datasets
- Removed duplicate records
- Filled missing values
- Standardized product names
- Converted date format
- Removed invalid quantity values
- Masked email and phone number for privacy
- Created revenue column

4. KPIs Generated

- Total Revenue
- Revenue by Category
- Revenue by City
- Total Orders
- Quantity Sold

5. Power BI Dashboard

The dashboard contains:

- KPI cards
- Revenue by category chart
- Revenue by city chart
- Product performance chart
- Revenue trend chart
- Payment method distribution

6. Conclusion

The retail dataset was cleaned and transformed successfully. Business insights were generated using a Power BI dashboard for decision making.